Experiment 25: On-Road Vehicle Breakdown Assistance System

Aim:

To develop an interactive website that helps users find licensed mechanics nearby when their vehicles break down. Users can search mechanic details, view location information, and post feedback. Mechanics can register and manage their business details, while administrators approve mechanics before their listings are shown.

Algorithm:

1. Start the application and display the home page.
2. Allow users to register, log in, and manage their profile.
3. Allow mechanics to register and submit their business and service details.
4. Allow the administrator to review and approve mechanic registrations.
5. Enable users to search approved mechanics by location and service type.
6. Display mechanic contact details and location information.
7. Allow users to submit feedback about the service received.
8. Allow mechanics to manage business information and view feedback.
9. Display the relevant information and end the session.

Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>RoadRescue | Vehicle Breakdown Assistance</title>

<link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css">

<style>

:root{--blue:#2563eb;--dark:#14213d;--muted:#64748b;--bg:#f4f7fc}

*{box-sizing:border-box}

body{margin:0;font-family:Arial,sans-serif;background:var(--bg);color:var(--dark)}
```

```
header{padding:20px 6%;background:#fff;display:flex;justify-content:space-between;align-items:center;gap:12px;flex-wrap:wrap;box-shadow:0 3px 15px #0000000b}

.logo{font-size:23px;font-weight:800;color:var(--blue)}

nav{display:flex;gap:7px;flex-wrap:wrap}

button{border:0;padding:11px 14px;border-radius:9px;font-weight:bold;cursor:pointer}

nav button{background:#edf2fa;color:#334155}

nav button.active,.primary{background:var(--blue);color:white}

main{max-width:1200px;margin:auto;padding:25px 5%}

.hero{padding:38px;border-radius:22px;color:white;background:linear-gradient(90deg,#0f1c38ed,#0f1c3835),url('https://images.unsplash.com/photo-1486262715619-67b85e0b08d3?auto=format&fit=crop&w=1500&q=85') center/cover}

.hero h1{font-size:clamp(30px,5vw,48px);max-width:620px;margin:0 0 12px}

.hero p{max-width:580px;line-height:1.7}

.stats{display:grid;grid-template-columns:repeat(3,1fr);gap:14px;margin:20px 0}

.stat,.panel,.mechanic{background:white;border-radius:15px;padding:20px;box-shadow:0 5px 20px #14213d0b}

.stat span{font-size:13px;color:var(--muted)}

.stat h2{color:var(--blue);margin:8px 0 0}

.grid{display:grid;grid-template-columns:repeat(2,minmax(0,1fr));gap:12px}

.field{margin-bottom:12px}

label{display:block;font-size:13px;font-weight:bold;margin-bottom:6px}

input,select,textarea{width:100%;padding:12px;border:1px solid #dbe2ed;border-radius:9px;font:inherit;background:white}

textarea{min-height:85px;resize:vertical}

.cards{display:grid;grid-template-columns:repeat(3,minmax(0,1fr));gap:15px}

.mechanic h3{margin:8px 0}

.tag{display:inline-block;background:#e8f1ff;color:#1d4ed8;padding:5px 9px;border-radius:20px;font-size:12px;font-weight:bold}

.muted{color:var(--muted);font-size:14px;line-height:1.5}
```

```
.hidden{display:none!important}
#map{height:340px;border-radius:14px;margin-top:14px;z-index:1}
.tablewrap{overflow-x:auto}
table{width:100%;border-collapse:collapse;min-width:570px}
th,td{text-align:left;padding:12px;border-bottom:1px solid #e5eaf2}
th{background:#eff4ff}
footer{text-align:center;padding:24px;color:var(--muted)}
@media(max-width:800px){.cards{grid-template-columns:repeat(2,minmax(0,1fr))}.hero{padding:25px}}
@media(max-width:550px){.cards,.grid,.stats{grid-template-columns:1fr}main{padding:18px 4%}}
</style>
</head>
<body>
<header>
 <div class="logo">🚙 RoadRescue</div>
 <nav>
  <button class="active" data-page="home" onclick="showPage('home',this)">Home</button>
  <button data-page="search" onclick="showPage('search',this)">Find Mechanic</button>
  <button data-page="register" onclick="showPage('register',this)">Mechanic Registration</button>
  <button data-page="feedback" onclick="showPage('feedback',this)">Feedback</button>
  <button data-page="admin" onclick="showPage('admin',this)">Admin</button>
 </nav>
</header>

<main>
```

```
<section id="home" class="page">
 <div class="hero">
  <span class="tag">ROAD ASSISTANCE DEMO</span>
  <h1>Vehicle trouble? Find help nearby.</h1>
  <p>Search mechanic listings, explore their locations, and find contact information when you need vehicle assistance.</p>
  <button class="primary" onclick="showPage('search')">Find a Mechanic →</button>
 </div>
 <div class="stats">
  <div class="stat"><span>Approved Mechanics</span><h2 id="totalApproved">0</h2></div>
  <div class="stat"><span>Service Areas</span><h2 id="totalAreas">0</h2></div>
  <div class="stat"><span>Customer Feedback</span><h2 id="totalFeedback">0</h2></div>
 </div>
 <div class="panel">
  <h2>How RoadRescue Works</h2>
  <div class="grid">
   <div><h3>🔎 Search</h3><p class="muted">Filter approved mechanic listings by location and service.</p></div>
   <div><h3>📍 View Location</h3><p class="muted">View available demo locations on the interactive map.</p></div>
   <div><h3>🛠️ Register</h3><p class="muted">Mechanics submit business details for administrator approval.</p></div>
   <div><h3>⭐ Feedback</h3><p class="muted">Submit feedback for a listed mechanic.</p></div>
  </div>
 </div>
</section>
```

```
<section id="search" class="page hidden">
 <h2>Find a Mechanic</h2>
 <div class="panel">
  <div class="grid">
   <div class="field"><label>Search location</label><input id="locationFilter" placeholder="e.g. Chennai" oninput="renderMechanics()"></div>
   <div class="field"><label>Service type</label>
    <select id="serviceFilter" onchange="renderMechanics()">
     <option value="">All services</option><option>Car Repair</option><option>Bike Repair</option>
     <option>Tyre Puncture</option><option>Battery Jumpstart</option><option>Towing</option>
    </select>
   </div>
  </div>
  <p class="muted">Only administrator-approved listings appear here.</p>
 </div>
 <div id="mechanicCards" class="cards"></div>
 <div class="panel"><h3>Mechanic Locations</h3><p class="muted">Map markers show demo business coordinates, not verified live availability.</p><div id="map"></div></div>
</section>

<section id="register" class="page hidden">
 <div class="panel">
  <h2>Mechanic Business Registration</h2>
  <p class="muted">New registrations remain pending until the administrator approves them.</p>
  <form id="mechanicForm">
   <div class="grid">
```

<label>Mechanic / Business Name</label><input name="name" required maxlength="80">

<label>Contact Number</label><input name="phone" required pattern="[0-9+ -]{8,16}">

<label>City / Area</label><input name="location" required maxlength="80">

<label>Service Type</label>

<select name="service" required>

<option>Car Repair</option><option>Bike Repair</option><option>Tyre Puncture</option>

<option>Battery Jumpstart</option><option>Towing</option>

</select>

<label>Business Address</label><input name="address" required maxlength="150">

<label>Latitude (optional)</label><input name="lat" type="number" step="any" min="-90" max="90" placeholder="13.0827">

<label>Longitude (optional)</label><input name="lng" type="number" step="any" min="-180" max="180" placeholder="80.2707">

<label>License / Registration Reference</label><input name="license" required maxlength="60">

<button class="primary">Submit for Approval</button>

</form>

<h3>My Business Listings</h3> Enter the registered contact number to view matching listing status. <input id="businessPhone" placeholder="Registered phone number" oninput="renderMyBusiness()">

</section>

<section id="feedback" class="page hidden">

<h2>Customer Feedback</h2>

<form id="feedbackForm">

<label>Approved Mechanic</label><select id="feedbackMechanic" required></select>

<label>Rating</label><select id="rating"><option value="5">★★★★★ — Excellent</option><option value="4">★★★★ — Very good</option><option value="3">★★★ — Good</option><option value="2">★★ — Fair</option><option value="1">★ — Poor</option></select>

<label>Your Feedback</label><textarea id="feedbackText" required maxlength="500" placeholder="Describe your experience"></textarea>

<button class="primary">Submit Feedback</button>

</form>

<h3>Recent Feedback</h3>

</section>

<section id="admin" class="page hidden">

<h2>Administrator Login</h2> Demo password: admin123

<label>Password</label><input id="adminPassword" type="password">

<button class="primary" onclick="loginAdmin()">Login</button>

<h2>Pending Mechanic Approvals</h2>

<h2>All Mechanic Listings</h2> <table>

<thead><tr><th>Business</th><th>Location</th><th>Service</th><th>Status</th><th>Action</th></tr></thead>

<tbody id="adminRows"></tbody>

</table> <button onclick="logoutAdmin()">Logout</button>

</section>

</main>

<footer>RoadRescue • Internet Programming Lab • Educational prototype</footer>

<script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"></script>

<script>

const MK="roadrescue_mechanics_v1",FK="roadrescue_feedback_v1";

const initial=[

{id:"M101",name:"City Auto Care",phone:"9000000011",location:"Chennai",service:"Car Repair",address:"Anna Salai, Chennai",license:"DEMO-LIC-101",status:"Approved",lat:13.0604,lng:80.2496},

{id:"M102",name:"Quick Bike Service",phone:"9000000012",location:"Chennai",service:"Bike Repair",address:"T Nagar, Chennai",license:"DEMO-LIC-102",status:"Approved",lat:13.0418,lng:80.2341},

{id:"M103",name:"Rapid Tyre Help",phone:"9000000013",location:"Coimbatore",service:"Tyre Puncture",address:"Gandhipuram, Coimbatore",license:"DEMO-LIC-103",status:"Approved",lat:11.0183,lng:76.9725}

];

function read(k,f=[]){try{return JSON.parse(localStorage.getItem(k))||f}catch{return f}}

function save(k,v){localStorage.setItem(k,JSON.stringify(v))}

let mechanics=read(MK);

if(!mechanics.length){mechanics=initial;save(MK,mechanics)}

let feedback=read(FK),admin=false,map=null,markers=null;

```
function esc(s){return String(s??"").replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;","'":"&quot;",'"':"&#39;"}[c]))}

function showPage(id,button){

 document.querySelectorAll(".page").forEach(x=>x.classList.add("hidden"));

 document.getElementById(id).classList.remove("hidden");

 document.querySelectorAll("nav button").forEach(x=>x.classList.remove("active"));

 const b=button||document.querySelector('nav button[data-page="'+id+'"]');if(b)b.classList.add("active");

 refresh();

 if(id==="search")setTimeout(initMap,100);

}

document.getElementById("mechanicForm").addEventListener("submit",e=>{

 e.preventDefault();const f=new FormData(e.target);

 const lat=f.get("lat")===""?null:Number(f.get("lat"));

 const lng=f.get("lng")===""?null:Number(f.get("lng"));

 if((lat===null)!==(lng===null)){alert("Enter both latitude and longitude, or leave both blank.");return}

 mechanics.push({

 id:"M"+Date.now().toString().slice(-8),name:f.get("name").trim(),

 phone:f.get("phone").trim(),location:f.get("location").trim(),

 service:f.get("service"),address:f.get("address").trim(),

 license:f.get("license").trim(),status:"Pending",

 lat:lat,lng:lng

 });

 save(MK,mechanics);e.target.reset();refresh();

 alert("Registration submitted. Status: Pending admin approval.");

});

function renderMechanics(){
```

```
 const q=document.getElementById("locationFilter").value.trim().toLowerCase();
 const service=document.getElementById("serviceFilter").value;
 const list=mechanics.filter(m=>m.status==="Approved"&&m.location.toLowerCase().includes(q)&&(!service||m.service===service));
 document.getElementById("mechanicCards").innerHTML=list.map(m=>
 `<article class="mechanic"><span class="tag">${esc(m.service)}</span><h3>${esc(m.name)}</h3>
 <p class="muted">📍 ${esc(m.address)}<br>Area: ${esc(m.location)}<br>☎ ${esc(m.phone)}</p>
 <button class="primary" onclick="showPage('feedback')">Give Feedback</button>
 ${m.lat!==null&&m.lng!==null?`<button onclick="focusMap(${Number(m.lat)},${Number(m.lng)})">View Map</button>`:""}
 </article>`).join("")||"<p>No approved mechanics match your search.</p>";
 renderMapMarkers(list);
}
function initMap(){
 if(typeof L==="undefined"){document.getElementById("map").innerHTML="Map library unavailable. Check your internet connection.";return}
 if(!map){
  map=L.map("map").setView([13.0827,80.2707],7);
  L.tileLayer("https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png",{attribution:"&copy; OpenStreetMap"}).addTo(map);
  markers=L.layerGroup().addTo(map);
 }
 map.invalidateSize();renderMapMarkers();
}
function renderMapMarkers(list){
 if(!map||!markers)return;
 markers.clearLayers();
```

```
 const shown=list||mechanics.filter(m=>m.status==="Approved");
 shown.forEach(m=>{

if(m.lat===null||m.lng===null||!Number.isFinite(Number(m.lat))||!Number.isFinite(Number(m.lng)))return;

L.marker([Number(m.lat),Number(m.lng)]).addTo(markers).bindPopup("<b>"+esc(m.name)+"</b><br>"+esc(m.address));
 });
}
function focusMap(lat,lng){showPage("search");setTimeout(()=>{if(map){map.setView([lat,lng],14);renderMapMarkers()}},150)}
function renderMyBusiness(){
 const phone=document.getElementById("businessPhone").value.trim();
 const found=mechanics.filter(m=>phone&&m.phone===phone);
 document.getElementById("myBusiness").innerHTML=found.map(m=>
 `<p>${esc(m.name)} — <span class="tag">${esc(m.status)}</span></p>`).join("")||"No matching listing found.";
}
function renderFeedback(){
 const approved=mechanics.filter(m=>m.status==="Approved");

document.getElementById("feedbackMechanic").innerHTML=approved.map(m=>`<option value="${esc(m.id)}">${esc(m.name)}</option>`).join("");

document.getElementById("feedbackList").innerHTML=feedback.slice().reverse().map(f=>{
 const m=mechanics.find(x=>x.id===f.mechanicId);
 return `<div class="panel"><b>${esc(m?m.name:"Mechanic listing")}</b> — ${"★".repeat(f.rating)}<p>${esc(f.text)}</p><span class="muted">${esc(f.date)}</span></div>`;
```

```
}).join("")||"<p>No feedback submitted yet.</p>";
}
document.getElementById("feedbackForm").addEventListener("submit",e=>{
 e.preventDefault();
 const id=document.getElementById("feedbackMechanic").value;
 if(!mechanics.some(m=>m.id===id&&m.status==="Approved")){alert("Select an approved mechanic.");return}

feedback.push({id:"F"+Date.now(),mechanicId:id,rating:Number(document.getElementById("rating").value),text:document.getElementById("feedbackText").value.trim(),date:new Date().toLocaleDateString()});

save(FK,feedback);document.getElementById("feedbackText").value="";refresh();alert("Feedback submitted.");
});
function loginAdmin(){
 if(document.getElementById("adminPassword").value==="admin123"){
  admin=true;document.getElementById("adminLogin").classList.add("hidden");
  document.getElementById("adminPanel").classList.remove("hidden");refresh();
 }else document.getElementById("loginMessage").textContent="Incorrect password.";
}
function logoutAdmin(){
 admin=false;document.getElementById("adminLogin").classList.remove("hidden");
 document.getElementById("adminPanel").classList.add("hidden");
 document.getElementById("adminPassword").value="";
}
function approve(id,status){
 if(!admin)return;
```

```
mechanics=mechanics.map(m=>m.id===id?{...m,status}:m);save(MK,mechanics);refr
esh();

}

function removeMechanic(id){

 if(!admin||!confirm("Delete this mechanic listing?"))return;

 mechanics=mechanics.filter(m=>m.id!==id);save(MK,mechanics);refresh();

}

function renderAdmin(){

 if(!admin)return;

 const pending=mechanics.filter(m=>m.status==="Pending");

 document.getElementById("pendingList").innerHTML=pending.map(m=>

 `<div class="panel"><b>${esc(m.name)}</b> — ${esc(m.service)}<p
class="muted">${esc(m.address)} | License reference: ${esc(m.license)}</p>

 <button class="primary"
onclick="approve('${esc(m.id)}','Approved')">Approve</button>

 <button
onclick="approve('${esc(m.id)}','Rejected')">Reject</button></div>`).join("")||"<p>No
pending registrations.</p>";

 document.getElementById("adminRows").innerHTML=mechanics.map(m=>


`<tr><td>${esc(m.name)}</td><td>${esc(m.location)}</td><td>${esc(m.service)}</td>
<td>${esc(m.status)}</td>

 <td><button
onclick="removeMechanic('${esc(m.id)}')">Delete</button></td></tr>`).join("");

}

function refresh(){

 mechanics=read(MK,initial);feedback=read(FK);


document.getElementById("totalApproved").textContent=mechanics.filter(m=>m.statu
s==="Approved").length;
```

```
 document.getElementById("totalAreas").textContent=new
Set(mechanics.filter(m=>m.status==="Approved").map(m=>m.location.toLowerCase())
).size;

 document.getElementById("totalFeedback").textContent=feedback.length;

 renderMechanics();renderMyBusiness();renderFeedback();renderAdmin();

}

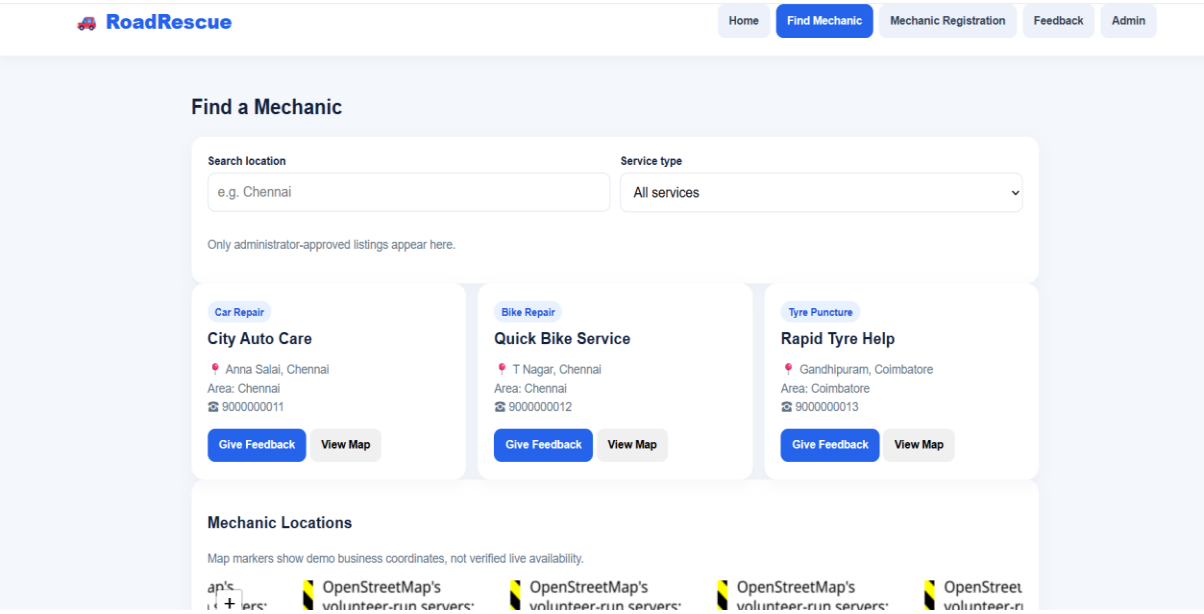
window.addEventListener("storage",refresh);refresh();

</script>

</body>

</html>
```

Output



Result

Thus, the On-Road Vehicle Breakdown Assistance System was developed using HTML, CSS, and JavaScript. It demonstrates mechanic registration, administrator approval, location-based searching, map display, and customer feedback.